

Capital Structure and Cost of Capital

Capital Structure

- **Capital structure** is the mix of debt and equity financing for a business enterprise.
- Capital structure can be fully equity financed, but there are not many business enterprises that will be fully financed by debt.
- Debt is a component which leverages itself on to the equity base.
- Outsiders are willing to lend to the business on the basis of the equity contribution.

Capital structure Decision

- Decision to finance part of the assets with borrowed funds has important managerial implications.
- If the firm finds it increasingly difficult to service its debt (paying interest and repaying the borrowed funds) because of excessive borrowing, its management will be under pressure to make decisions that may not be in the best interest of shareholders.

Example

- Management may have to quickly sell value-creating assets for less than they are worth in order to raise the cash needed to make the payments required to service the business debt.
- Conversely, a business with too little debt may pass up the opportunity to reduce its tax payments and increase its value through tax savings.
- By replacing equity with debt, the firm can deduct the interest expenses from its taxable income and save the cash that would have been used to pay taxes.

Capital Structure in Practice

- Managers can choose from a variety of sources of funds to finance their business.
- Most of these are hybrids of two basic types of capital: debt, such as bank loans and bonds, and equity, which includes retained earnings and common stocks.
- How managers should combine debt and equity financing to establish a capital structure that maximizes the value of the firm's assets is a pertinent question for the financial manager.
- Capital structure is usually identified by its debt ratios, either its debt-to-equity ratio (the amount of borrowing divided by the amount of equity) or its debt-to-assets ratio (the portion of the assets financed with borrowed funds). These two debt ratios are often used interchangeably.

Weighted Average Cost of Capital

- Weighted average cost of capital is the average of the costs of the firm's debt and equity weighted by their proportions in the firm's capital structure.

$$r^* = r_D (1 - T_c) \frac{D}{V} + r_E \frac{E}{V}$$

r^* = The adjusted cost of capital

r_D = The firm's current borrowing rate

T_c = The marginal corporate income tax rate

r_E = The expected rate of return on the firm's stock

D, E = The market values of currently outstanding debt and equity

$V = D + E$ = The total market value of the firm.

- The weights in the above formula is based on the market values. However, a firm may also use the book value of debt and equity in its financial composition to compute the weighted average cost of capital.

WEIGHTED AVERAGE COST OF CAPITAL

Example:

Arise limited wishes to raise additional finance of INR 10 lakhs for meeting its investment plans. It has INR 2,10,000 in the form of retained earning available for investment purpose. The following are the further details.

1) Debt / equity mix	30% / 70%
2) Cost of debt upto Rs 1,80,000 beyond Rs.1,80,000	10% (before tax) 16% (before tax)
3) Earnings per share	Rs.4
4) Dividend pay out	50% of earnings
5) Expected growth rate in dividend	10%
6) Current market price per share	Rs.44
7) Tax rate	50%

Requirement :

- a) Determine the pattern for raising the additional finance.
- b) Determine the post-tax average cost of additional debt.
- c) Determine the cost of retained earnings and cost of equity.
- d) Compute the weighted average after tax cost of additional finance.

a)Pattern for raising the additional finance:-

Debt 30% of 10 lakhs INR 3,00,000

Equity 70% of 10 lakhs	<u>INR 7,00,000</u>
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Total	INR10,00,000
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<i>Source</i>	<i>Amount</i>	<i>Cost</i>
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Debt	1,80,000	10%
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Debt	<u>1,20,000</u>	16%
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3,00,000

Retained Earnings	2,10,000
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Equity 7,00,000-2,10,000	<u>4,90,000</u>
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Total	10,00,000
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B)Post - tax average cost of additional debt:-

Formula : $kd(1-r)$

$$kd = 1,80,000 \times 10\% = 18,000$$

$$\text{Balance} = \underline{1,20,000} \times 16\% = \underline{19,200}$$

$$\text{Total} = \underline{3,00,000} \quad 37,200$$

$$37,200 / 3,00,000 \times 100 = 12.4\%$$

$$\text{Post-tax average cost of debt} = 12.4(1-0.5) = 6.2\%$$

C) Cost of Retained Earnings and cost of equity:

Cost of Retained Earnings:-

$$K_r = (d_i / p_o) + g$$

$$k_r = (2.20 / 44) + 0.10 = 0.05 + 0.10 \text{ i.e. } 15\% \text{ (Note 1)}$$

D) Cost of Equity:

$$k_e = (2.20 / 44) + 0.10 = 0.05 + 0.10 \text{ i.e. } 15\%$$

Note 1:

Dividend pay-out = 50% of earnings i.e. 50% of 4 = INR 2 per share

Growth rate g is 10%. Hence, at the end of year, dividend will be = INR 2 + 10%
= 2.20

(d) Weighted Average after tax cost of additional finance:-

Sources of finance	Amount	Proportion	After-tax cost	Weighted Average cost(%)
Debt	3,00,000	30%	6.2%	$30\% \times 6.2\% = 1.86\%$
Retained Earnings	2,10,000	21%	15%	$21\% \times 15\% = 3.15\%$
Equity	<u>4,90,000</u>	<u>49%</u>	15%	$49\% \times 15\% = \underline{7.35\%}$
Total	10,00,000	100%		12.36%

Division / Project Cost of Capital

- Division/Project cost of capital is the risk-adjusted discount rate at which the individual project's cash flow should be discounted. When a firm evaluated individual projects, the project discount rate may vary from one project to another because risk may vary.
- Risk component varies due to the composition of project finance.
- A project that is fully financed by equity is called an unlevered project and one that is partially financed by equity and partially by debt is called a levered project.

Example

M/s Efficient Corporation has a capital structure of 40% debt and 60% equity. The company is presently considering several alternative investment proposals costing less than INR 20 million. The corporation always raises the required funds without disturbing its present debt equity ratio. The cost of raising the debt and equity are as under:

	cost of debt	cost of equity
Upto INR 2 million	10%	12%
Above INR 2 million & upto INR 5 million	11%	13%
Above INR 5 million & upto INR 10 million	12%	14%
Above INR 10 million & upto INR 20 million	13%	14.5%

Assuming the tax rate at 50% calculate:-

- (i) Cost of capital of two projects X and Y whose fund requirements are INR 6.5 million and INR 14 million respectively.
- (ii) If a project is expected to give after tax return of 10% determine under what conditions it would be acceptable?

Alternate Capital Structures

Project cost	Financing	Proportion of capital structure	After tax cost (1-Tax 50%)	Weighted average cost(%)
Upto INR 2 m	Debt	0.4	10%(1-5)=5%	0.4 X 5 = 2.0
	Equity	0.6	12%	0.6 X 12= <u>7.2</u> 9.2%
Above INR 2 m & upto INR 5 m	Debt	0.4	11%(1-5)=5.5%	0.4 X 5.5 = 2.2
	Equity	0.6	13%	0.6 X 13= <u>7.8</u> 10.0%
Above INR 5 m & upto INR 10 m	Debt	0.4	12%(1-5)=6%	0.4 X 6 = 2.0
	Equity	0.6	14%	0.6 X 14= <u>8.4</u> 10.8%
Above INR 10 m & upto INR 20 m	Debt	0.4	13%(1-5)=6.5%	0.4 X 6.5 = 2.6
	Equity	0.6	14.5%	0.6 X 14.5= <u>8.7</u> 11.3%

Project Selection

<i>Project</i>	<i>Fund requirement</i>	<i>cost of capital</i>
X	INR.6.5 million	10.8%
Y	INR 14 million	11.3%

(ii) if a project is expected to give after tax return of 10%, it would be acceptable provided cost does not exceed INR 5 million.

Acceptance criterion:

After tax return should be more than or at least equal to the weighted average cost of capital.

Adjusted Cost of Capital

- All variables in the weighted average cost of capital (WACC) formula refer to the firm as a whole.

$$WACC = r_D(1 - T_C)\left(\frac{D}{V}\right) + r_E\left(\frac{E}{V}\right)$$

- Where T_C is the corporate tax rate.
- After-tax WACC can be used as the discount rate if
 - Project has the same business risk as the average project of the firm (1)
 - Project is financed with the same amount of debt and equity (2)
 - If condition 1 is violated the right discount factor is the required rate of return on an equivalently risky investment,
 - if condition 2 is violated the WACC should be adjusted to the right financing mix.

Opportunity Cost of Capital

- Calculate the opportunity cost of capital without corporate taxation

$$r = \frac{D}{V}r_D + \frac{E}{V}r_E$$

- Estimate the cost of debt, r_D , and cost of equity, r_E , at the new debt level.
- Recalculate WACC with the new financing weights. (adjustment to be made)

$$r_E = r + (r - r_D)\frac{D}{E}$$

Example 1

- Consider a firm with a debt and equity ratio of 40% and 60%, respectively. Required rate of return on debt and equity is 7% and 12.5%, respectively. Assuming a 30% corporate tax rate the after-tax WACC of the firm is:

$$7\% \times (1-0.3) \times 0.4 + 12.5\% \times 0.6 = 9.46\%$$

- Firm is considering investing in a new project with a perpetual stream of cash flows of USD 11.83 million per year pre-tax. Project has the same risk as the average project of the firm.
- Given an initial investment of USD 125 million, which is financed with 20% debt, the value of the project would be:

Example 1: Opportunity Cost

- Business risk is similar but the project is financed with lower financial leverage. Hence the WACC of 9.46% cannot be used as the discount rate for the project.
- WACC should be adjusted
- Estimate opportunity cost of capital, (estimate r using a 40% debt ratio, 60% equity ratio as well as the firm's cost of debt and equity)

$$7\% \times 0.4 + 12.5\% \times 0.6 = 10.3\%$$

Example 1: Project Cost

- Estimate the expected rate of return on equity using the project's debt-equity ratio. As the debt ratio is equal to 20%, the debt-equity ratio equals (20/80) 25%.

$$10.3\% + (10.3\% - 7\%) \times 0.25 = 11.1\%$$

- Estimate the project's WACC

$$7\% \times (1 - 0.3) \times 0.2 + 11.1\% \times 0.8 = 9.86\%$$

Example 1 : Project Present Value

- Adjusted WACC of 9.86% can be used as the discount rate for the new project as it reflects the underlying business risk and mix of financing. As the project requires an initial investment of USD 125 million and produced a constant cash flow of USD 11.83 per year for ever,
- Projects NPV is:
$$NPV = -125 + \frac{11.83}{0.0986} = - \text{USD } 5.02 \text{ million}$$

Example 1

- In comparison the NPV is equal to USD 5.03 if the company WACC is used as the discount rate.
- Firm could have invested in a negative NPV project if it had ignored that the project was financed with a different mix of debt and equity.

Example 2

- A firm has a market value of INR 200 million and its 6% bonds are valued at par (INR 50 million). Firm's beta is 1.3.
- Firm is worried about the recession and is considering diversification. It finds firm DF could closely match its plans. DF has a beta of 0.9 and the ratio of the market value of its equity to its debt is 7:5.
- Firm plans to finance its new venture with a market value of equity to debt ratio of 1:1
- Corporate tax rate is 20%. Risk free rate is 5.5%. Market return is 17.5%
- What discount rate should be used to evaluate the new venture?

Cost of Equity (CAPM)

- $K_e = r_f + (r_m - r_f) \times \beta_i$
- $5.5\% + (17.5\% - 5.5\%) \times 1.3 = 21.1\%$

Equity Beta Vs Asset Beta

- Beta measured by the market is the Equity beta.

$$\beta_{\text{equity}} = \beta_{\text{asset}} \left[1 + (1 - \text{Tax rate}) \frac{\text{Debt}}{\text{Equity}} \right]$$

$$\beta_{\text{asset}} = \frac{\beta_{\text{equity}}}{\left[1 + (1 - \text{Tax rate}) \frac{\text{Debt}}{\text{Equity}} \right]}$$

- Equity beta of DF is 0.9.
- Hence Asset beta of DF = $0.9 / [1 + (1 - 0.2) \times (5/7)]$
- = $0.9 / 1.5714 = 0.5727$

Adjusting the beta for debt equity

- Asset beta (un-gearred beta) can be adjusted to reflect the gearing ratio planned in the new venture.
- $0.5727 = (1 / [1+(1-0.2)]) \times \text{Equity beta}$
- $= 0.5727 \times 1.8 = 1.03$
- Cost of Equity of DF = $5.5 + 0.5727 (17.5 - 5.5)$
- $= 12.37\%$

Cost of equity and cost of debt

- Firm is planning for a higher leverage ratio of 1 : 1
- Required rate of return = $5.5 + (1.03 (17.5 - 5.5)) = 17.86\%$
- Cost of equity of the shareholders would be 17.86%
- Cost of debt 6%.
- After tax cost of debt = $6\% \times (1 - 0.2) = 4.8\%$
- WACC = $(17.86\% \times 0.5) + (4.8\% \times 0.5) = 11.33\%$

Weighted Average Cost of Capital

- Change in D/E Implication

